

Cooperative Perception

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I. SYSTEM MODEL

A. Scenario Description

We consider an edge-coordinated, V2V-executed, and V2I-assisted cooperative perception scenario in an urban driving environment, as illustrated in Fig. 1. A base station (BS), connected to an edge controller, covers a local road segment and serves nearby connected autonomous vehicles (CAVs). The system is time-slotted with slot length T_s , which serves as the sole decision time unit throughout the paper; all cadence, window, and duration quantities are expressed as integer numbers of slots. When a CAV enters the coverage area of the BS, it registers as an expert agent at the controller and delegates its cooperative-perception decisions to the controller for the duration of its dwell: at every instant, the controller determines which cooperative-perception action the vehicle executes, with purely local sensing (*local-only*) being one action among others. The controller maintains a lightweight world model for each registered CAV based on the vehicle profile, mobility history, and perception summary. The world model comprises an environment expert and an ego expert, used to predict the evolution of the surrounding environment and the ego state under a candidate cooperative-perception action chunk. The controller periodically searches over candidate action chunks based on the world model—each chunk specifying not only *what* to execute but also *for how long*—and delivers the selected chunk to the vehicle via a V2I control link; the resulting decision rule constitutes the cooperation *policy*. Each action determines, for example, which nearby vehicle should collaborate with the requesting vehicle. The vehicles selected by the controller, referred to as *member* vehicles, transmit task-relevant sensing information directly to the target vehicle through V2V data links, following the issued action. In the considered scenario, multiple CAVs may be managed by the BS simultaneously. Let $\mathcal{V}_t = \{1, \dots, V_t\}$ denote the set of CAVs within the coverage of the BS at time slot t . Since the decision-making process is identical for each managed CAV, we focus on a single representative CAV without loss of generality, referred to as the target vehicle v . After receiving sensing information from the selected member vehicles, the target vehicle fuses the shared information with its locally captured data to achieve cooperative perception, and reports lightweight execution feedback to the BS. To ensure the continued effectiveness of the issued chunk, the controller compares the realized perception performance against the performance predicted at decision time. If a significant gap appears between the expected performance and the actual

observed performance, an online adjustment is triggered, and the controller conducts a new chunk search (Sec. ??).

B. Cooperative Perception Action, Action Chunk, and Policy

1) *Action*: A cooperative perception action is the atomic decision unit issued by the BS for the target vehicle v :

$$a = (\mathcal{S}, \mathcal{B}, \mathcal{D}, n), \quad (1)$$

where $\mathcal{S}$ is the selected member set, $\mathcal{B} = \{B_m\}_{m \in \mathcal{S}}$ the per-member bandwidth allocation, $\mathcal{D} = \{d_m\}_{m \in \mathcal{S}}$ the transmitted modalities with $d_m \subseteq \{\text{objlist}, \text{bevfeature}\}$, and n the *execution duration* of the action, expressed as an integer number of slots. Durations satisfy

$$n_{\min} \leq n, \quad (2)$$

where the lower bound $n_{\min}$ reflects the overhead of V2V link establishment and member negotiation, which precludes slot-level reconfiguration. The local-only action is $a^{\text{loc}}(n) = (\emptyset, 0, \emptyset, n)$. Each action is thus temporally extended in the sense of semi-Markov decision processes, with a duration chosen rather than fixed.

2) *Action chunk*: At each decision epoch t_k , rather than committing to a single action, the controller issues an *action chunk* [2]: an ordered sequence of J actions

$$\mathbf{a}_{t_k} = (a_{t_k}^{(0)}, a_{t_k}^{(1)}, \dots, a_{t_k}^{(J-1)}). \quad (3)$$

With the sub-epoch start times

$$t_k^{(j)} \triangleq t_k + \sum_{i < j} n^{(i)}, \quad j = 0, \dots, J, \quad (4)$$

the j -th *sub-action* governs the j -th *sub-epoch*

$$\mathcal{E}_k^{(j)} = [t_k^{(j)}, t_k^{(j+1)}], \quad j = 0, \dots, J-1, \quad (5)$$

so that the slot-wise active action is piecewise constant,

$$a_t = a_{t_k}^{(j)}, \quad \forall t \in \mathcal{E}_k^{(j)}. \quad (6)$$

The *planned horizon* of the chunk is its total duration

$$F(\mathbf{a}_{t_k}) = \sum_{j=0}^{J-1} n^{(j)} \leq F_{\max}, \quad (7)$$

where the cap $F_{\max}$ explicitly bounds the open-loop commitment, and $t_k^{(J)} = t_k + F(\mathbf{a}_{t_k})$.

When the chunk $\mathbf{a}_{t_k}$ is installed, the world-action model simultaneously produces and stores the reference uncertainty trajectory $\{\hat{U}_{v,t|t_k}\}_{t \in [t_k, t_k + F(\mathbf{a}_{t_k})]}$, i.e., the per-slot uncertainty that the chunk is predicted to achieve. At run time

the vehicle computes the realized $U_{v,t}(a_t)$ every monitoring period T_a and reports its deviation from the reference to the BS. Re-planning is invoked only when the gap between reality and expected value is more than a mismatch tolerance $\epsilon_{\text{gap}} > 0$. The next decision epoch is

$$t_{k+1} = \min \left\{ t_k + F(\mathbf{a}_{t_k}), \right. \\ \left. \inf \left\{ t > t_k + T_{\min} : |U_{v,t}(a_t) - \hat{U}_{v,t|t_k}| > \epsilon_{\text{gap}} \right\} \right\}. \quad (8)$$

3) *Policy*: A *policy* π is a decision rule that maps the information available at each decision epoch, namely the global context $\mathcal{C}_{v,t_k}^{\text{BS}}$ maintained by the controller and the queue state $\Theta(t_k)$ (Sec. IV), to an action chunk:

$$\pi : (\mathcal{C}_{v,t_k}^{\text{BS}}, \Theta(t_k)) \mapsto \mathbf{a}_{t_k}. \quad (9)$$

Setting $J = 1$ recovers a single-temporally-extended-action-per-epoch scheme as a special case.

C. Semantic Uncertainty

At each time slot, the reliability of the vehicle's environmental understanding is assessed with respect to the current driving task. Rather than relying on generic detection confidence, we define semantic uncertainty as a task-oriented and region-aware measure. Specifically, it emphasizes objects that are critical to the intended maneuver, referred to as *notable* objects, as well as the spatial region that the ego vehicle is expected to traverse, referred to as the *route corridor*. The resulting uncertainty is decomposed into two complementary components: predictive motion uncertainty over task-relevant objects and spatial coverage uncertainty over the route corridor. Throughout this subsection, the superscript a denotes quantities induced by the currently active action a_t in (6).

1) *Predictive motion uncertainty*: Given a fusion function $\Psi_{v,t}^a$, it fuses sensing data from multiple CAVs to predict the motion of each perceived object $o \in \mathcal{O}_t^a$ at future step $\tau \in \{1, \dots, H\}$. A diagonal-covariance distribution $\mathcal{N}(\boldsymbol{\mu}_{o,t+\tau}^a, \Sigma_{o,t+\tau}^a)$ with $\Sigma_{o,t+\tau}^a = \text{diag}(\sigma_{x,o,t+\tau}^{2,a}, \sigma_{y,o,t+\tau}^{2,a})$ represents the vehicle's ability to understand the environment under the active action. The predictive uncertainty of object o is the horizon-averaged trace of its covariance,

$$\text{Tr } \Sigma_{o,t}^a = \frac{1}{H} \sum_{\tau=1}^H (\sigma_{x,o,t+\tau}^{2,a} + \sigma_{y,o,t+\tau}^{2,a}). \quad (10)$$

Because only task-relevant objects should drive cooperation, each object is weighted by a predicted notable probability $w_{o,t}$. The motion uncertainty is the notable-weighted mean trace,

$$\tilde{U}_{v,t}^{\text{mot}}(a) = \frac{\sum_{o \in \mathcal{O}_t^a} w_{o,t} \text{Tr } \Sigma_{o,t}^a}{\sum_{o \in \mathcal{O}_t^a} w_{o,t} + w_0}, \quad (11)$$

where the constant $w_0 > 0$ is a notable-mass floor that keeps (11) well behaved when no confidently notable object is present ($\sum_o w_{o,t} \ll w_0$). The motion uncertainty is then normalized to $[0, 1]$ by

$$U_{v,t}^{\text{mot}}(a) = 1 - e^{-\tilde{U}_{v,t}^{\text{mot}}(a)/\sigma_0}, \quad (12)$$

where σ_0 is a predefined parameter serving as a reference scale for variance.

2) *Spatial coverage uncertainty*: Predictive uncertainty only accounts for objects that are already perceived; it cannot express the risk of unobserved regions along the intended path. Let $\mathcal{R}_{v,t}$ be the sub-regions of the target vehicle's route corridor. Each sub-region r carries a proximity-based route risk

$$\rho_{r,t} = \frac{1}{1 + d(r, v)/d_0}, \quad (13)$$

so that sub-regions nearer the target vehicle matter more. The coverage quality of sub-region r under the active action a is the best quality among the target vehicle and the selected members $\mathcal{S}_t$,

$$\chi_{r,t}^a = \max_{v' \in \{v\} \cup \mathcal{S}_t} \mathbb{1} \left[r \in \text{FoV}_{v',t} \wedge \text{LoS}(v', r) \right] \frac{\beta_{v',t}}{1 + d(r, v')/d_1}, \quad (14)$$

where a sub-region contributes only if it lies in the observer's field of view (FoV) and its line of sight (LoS) is unoccluded, the quality decays with the observer-sub-region distance, and $\beta_{v',t}$ is a freshness factor that discounts stale cooperative observations, $\beta_{v',t} = e^{-\gamma \zeta_{v',t}}$ with message age $\zeta_{v',t}$, and $\beta_{v',t} = 1$ for the target vehicle's own current sensing. The coverage uncertainty is the route-risk-weighted fraction of the corridor that remains poorly observed,

$$U_{v,t}^{\text{cov}}(a) = \frac{\sum_{r \in \mathcal{R}_{v,t}} \rho_{r,t} (1 - \chi_{r,t}^a)}{\sum_{r \in \mathcal{R}_{v,t}} \rho_{r,t}} \in [0, 1]. \quad (15)$$

3) *Task-oriented semantic uncertainty*: The semantic uncertainty of vehicle v under the active action a aggregates the two terms,

$$U_{v,t}(a) = \alpha U_{v,t}^{\text{mot}}(a) + (1 - \alpha) U_{v,t}^{\text{cov}}(a), \quad (16)$$

with weight α balancing object motion prediction risk against region blind-spot risk. By construction of (12), (15), and the convex combination (16), the semantic uncertainty is bounded, $U_{v,t}(a) \leq 1$.

The realized $U_{v,t}(a_t)$ is computed on the vehicle at run time and plays two roles in the cooperative-perception loop: it is the per-slot evaluation reward of candidate chunks during the policy search, and it is the execution feedback whose deviation from the plan drives re-planning.

D. Communication Model

We model V2V cooperative perception as an action-conditioned streaming process. The BS issues the target vehicle v an action chunk, and the selected member vehicles continuously stream their observations to v . The active action and the induced communication latency jointly determine which historical observations are available to v at each time slot. The latency components below are accumulated in continuous time and discretized once at delivery.

While a sub-action is active, each member $m \in \mathcal{S}_t$ obtains its sensor data every slot and emits a message carrying all of its modalities. Its payload size is

$$L_{m,t}^{a_t} = L_0 + \sum_{d \in \mathcal{D}_{m,t}^{a_t}} L_{m,t}^d, \quad (17)$$

where L_0 represents a fixed overhead.

1) *Message timeline and queuing delay*: We index the i -th message emitted on the link from member m to the target vehicle by the pair (m, i) , and denote its payload by $L_{m,i}$. Each message carries four timestamps. It is sensed at $T_{m,i}^{\text{sense}}$ and becomes ready to enter the transmit queue after a processing delay T^{proc} :

$$T_{m,i}^{\text{ready}} = T_{m,i}^{\text{sense}} + T^{\text{proc}}. \quad (18)$$

Each member–target link maintains a FIFO transmit queue. If the link is idle the message is sent immediately; otherwise it waits behind the in-flight message, so

$$T_{m,i}^{\text{send}} = \max(T_{m,i}^{\text{ready}}, T_{m,i-1}^{\text{recv}}), \quad (19)$$

$$T_{m,i}^{\text{queue}} = T_{m,i}^{\text{send}} - T_{m,i}^{\text{ready}}, \quad (20)$$

where $T_{m,i-1}^{\text{ready}}$ is the time when the previous message is received.

$$\begin{aligned} T_{m,i}^{\text{recv}} &= T_{m,i}^{\text{send}} + T_{m,i}^{\text{tx}}, \\ T_{m,i}^{\text{tx}} &= \frac{L_{m,i}}{R_{m,v,t}}, \quad t_m^{\text{busy}} \end{aligned} \quad (21)$$

where $T_{m,i}^{\text{tx}}$ is transmission time. The end-to-end latency decomposes as

$$\Gamma_{m,i} = T_{m,i}^{\text{recv}} - T_{m,i}^{\text{sense}} = T^{\text{proc}} + T_{m,i}^{\text{queue}} + T_{m,i}^{\text{tx}}. \quad (22)$$

Since members stream every slot while a link drains one message in $T^{\text{proc}} + T_{m,i}^{\text{tx}}$, a backlog accumulates whenever

$$T^{\text{proc}} + T_{m,i}^{\text{tx}} > T_s, \quad (23)$$

i.e., when the per-message service time exceeds the streaming period; otherwise the queue stays empty and $T_{m,i}^{\text{send}} = T_{m,i}^{\text{ready}}$.

2) *Transmission rate*: The link rate follows the Shannon capacity under a large-scale path-loss channel. With transmit power P^{tx} and member–target vehicle distance $d_{m,v}$,

$$P_{m,v}^{\text{rx}} = P^{\text{tx}} - PL(d_{m,v}), \quad (24)$$

$$PL(d_{m,v}) = 38.77 + 16.7 \log_{10}(d_{m,v}) + 18.2 \log_{10}(f_c), \quad (25)$$

$$R_{m,v,t} = B_{m,t} \log_2 \left(1 + \frac{P_{m,v}^{\text{rx}}}{N_0 B_{m,t}} \right), \quad (26)$$

where f_c is the carrier frequency, N_0 the noise power spectral density, and $B_{m,t}$ the allocated bandwidth. Through (26), selecting richer modalities or allocating less bandwidth (smaller $B_{m,t}$) lengthens $T_{m,i}^{\text{tx}}$, which via (22)–(23) raises the latency and can destabilize the queue.

3) *Queue stability*: To prevent the backlog in (23) from growing unboundedly—which would push the latency $\Gamma_{m,i}$ beyond the prediction window T_w and effectively reduce cooperation to local-only perception—we stabilize the transmit queues with Lyapunov optimization. We model each member–target link as a bit queue with backlog $Q_m(t)$,

$$Q_m(t+1) = \max\{Q_m(t) - R_{m,t} T_s, 0\} + \mathcal{K}\{m \in \mathcal{S}_t\} L_{m,t}^{a_t}, \quad (27)$$

where the arrival $\mathcal{K}\{m \in \mathcal{S}_t\} L_{m,t}^{a_t}$ is the payload (17) injected when member m is selected, and $R_{m,t} T_s$ is the number of bits served in one slot. The backlog is a queuing-delay proxy, $T_{m,i}^{\text{queue}} \approx Q_m(t)/R_{m,t}$. For compactness, we write the per-slot arrival as

$$L_m(t) \triangleq \mathcal{K}\{m \in \mathcal{S}_t\} L_{m,t}^{a_t}. \quad (28)$$

Since payloads, bandwidths, and rates are bounded, $L_m(t) \leq L_{\max}$ and $R_{m,t} \leq R_{\max}$; we let $\delta_{\max} \triangleq \max\{L_{\max}, R_{\max} T_s\}$ denote the maximum per-slot queue change.

E. Uncertainty-Based Controller

The target vehicle follows a fixed reference route and regulates only its longitudinal speed, which it adapts to the prevailing semantic uncertainty through a risk-aware controller. This closes a loop between perception and motion: an action that lowers uncertainty lets the vehicle proceed faster, whereas residual uncertainty forces it to slow.

At slot t the controller evaluates a set of acceleration candidates $u \in \mathcal{U}^{\text{ctl}}$. Each candidate induces a short-horizon speed profile along the reference route,

$$\nu_{t+\tau}(u) = \text{clip}(\nu_t + u \tau \Delta t, 0, \nu_{\max}), \quad \tau = 1, \dots, H, \quad (29)$$

where ν_t is the current longitudinal speed and Δt the physical integration step of the motion model, and a corresponding pose $\mathbf{p}_{t+\tau}(u)$ obtained by advancing along the route by the arc length $\sum_{i=1}^{\tau} \nu_{t+i}(u) \Delta t$. The predictive uncertainty of each notable object inflates the radius within which it must be avoided,

$$r_{o,t+\tau}^{\text{eff}} = r_o + \kappa \sqrt{\text{Tr} \Sigma_{o,t+\tau}^a}, \quad (30)$$

so that a less certain object (larger trace) occupies a larger danger region. The risk of a candidate is the notable-weighted proximity of its trajectory to the predicted objects,

$$\text{Risk}_t(u) = \sum_{\tau=1}^H \sum_{o \in \mathcal{O}_t^a} w_{o,t} \exp \left(- \frac{\|\mathbf{p}_{t+\tau}(u) - \boldsymbol{\mu}_{o,t+\tau}^a\|^2}{2(r_v + r_{o,t+\tau}^{\text{eff}} + \epsilon_{\text{sf}})^2} \right), \quad (31)$$

with target-vehicle radius r_v and safety margin ϵ_{sf} . The controller trades progress against risk and control effort,

$$u_t^* = \arg \max_{u \in \mathcal{U}^{\text{ctl}}} \omega_\nu \bar{\nu}_t(u) - \omega_r \text{Risk}_t(u) - \omega_u |u|, \quad (32)$$

$$\bar{\nu}_t(u) = \frac{1}{H} \sum_{\tau} \nu_{t+\tau}(u),$$

and applies u_t^* , advancing v to the next pose. The ego state is thus *endogenous* to the action sequence: the issued actions

shape not only what the vehicle observes but also where it will be, and therefore the uncertainty it accrues along the way. An important consequence of (30)–(32) is a clear division of responsibility between the two layers: residual uncertainty that cooperation cannot eliminate is absorbed by the motion layer as increased conservatism (larger avoidance radii and lower speed), so safety is guaranteed by the controller under *any* residual uncertainty, while cooperative perception targets driving *efficiency* by restoring the confidence required to proceed at speed.

II. PROBLEM FORMULATION

Building on the system model, we formulate the cooperative-perception decision problem for the target vehicle v over a long horizon. From the moment the vehicle registers, the controller is responsible for its action sequence; since re-deciding at every slot is neither computationally nor communication-wise affordable, the controller commits, at sparse decision epochs, to an action chunk, and revises the plan only when the realized perception quality deviates from what was predicted. We seek a policy that minimizes the target’s long-term semantic uncertainty together with the re-planning overhead, **subject to a long-term communication-resource budget**.

A. Formulation

Let $K(T)$ denote the number of decision epochs in $[0, T)$. Each epoch incurs a fixed re-planning overhead $c_0 \geq 0$, representing the computation of one world-model chunk search plus the V2I signaling required to deliver the chunk; c_0 is expressed in the units of the uncertainty cost and **can be calibrated from the measured cost of one planning round**. To minimize two long-term quantities for the target vehicle: the time-average semantic uncertainty and the time-average allocated communication bandwidth,

$$\begin{aligned}\bar{U}_c &= \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} U_{v,t}(a_t) + c_0 K(T) \right], \\ \bar{B} &= \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left[\sum_{m \in \mathcal{S}_t} B_{m,t} \right],\end{aligned}\quad (33)$$

where the first term of $\bar{U}_c$ reflects how reliably the target understands its task-relevant environment, the second penalizes the re-planning rate, and $\bar{B}$ is the spectrum consumed for cooperation. Including c_0 makes the decision cadence part of the optimization: a policy that re-plans too often pays the overhead, while one that commits too long forgoes adaptation—precisely the trade-off that the choice of durations must navigate. Perception and communication remain coupled: reducing $\bar{B}$ lowers the link rate (26) and raises latency, whose staleness discounts the shared observations and in turn increases $\bar{U}_c$.

We impose a *long-term bandwidth budget* $\bar{B}_{\text{bgt}} > 0$, a physically meaningful quantity (in Hz) that can be directly specified by spectrum management, e.g., as the average share of cell bandwidth granted to the cooperative-perception service. To

simplify the problem, we allow at most one collaborating CAV to assist the target vehicle at a time, transmitting BEV features only. The cooperative-perception policy optimization is formulated as

$$(\mathbf{P1}) \quad \min_{\pi} \quad \bar{U}_c \quad (34a)$$

$$\text{s.t.} \quad (2)–(7), (8) \quad (34b)$$

$$|\mathcal{S}_t| \leq 1, \quad \forall t \quad (34c)$$

$$0 \leq \sum_{m \in \mathcal{S}_t} B_{m,t} \leq B_{\text{max}}, \quad \forall t \quad (34d)$$

$$\bar{B} \leq \bar{B}_{\text{bgt}} \quad (34e)$$

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[Q_m(t)] < \infty, \quad \forall m. \quad (34f)$$

Constraint (34b) encodes the action space with its duration structure, the chunk composition, and the event-driven epoch dynamics. Constraint (34c) allows the target at most one member at a time. Constraint (34d) is the *instantaneous* cap imposed by the physical spectrum available at any slot, whereas (34e) is the *long-term average* budget; the two are complementary and neither implies the other. Constraint (34f) enforces mean-rate stability of every member transmit queue (27).

B. Challenges

Solving **(P1)** is challenging because it involves online optimization over a sequence of coupled discrete–continuous action chunks under mobility and communication dynamics. First, the objective and the constraints (34e)–(34f) are coupled across slots through the queue dynamics (27) and the cumulative bandwidth consumption. A myopic rule that minimizes the current uncertainty may accumulate excessive backlog or exhaust the long-term budget. Second, both the wireless channel and vehicle mobility are time-varying and unknown a priori. Since they jointly determine the transmission rate in (26) and the resulting semantic uncertainty in (16), the system cannot rely on an offline solution computed from future information. Third, because cooperation modulates the vehicle’s own motion through the uncertainty-coupled controller (32), a candidate chunk cannot be scored by its instantaneous cost alone: its effect on future uncertainty propagates through the induced ego trajectory. Finally, the chunk decision space—member, bandwidth, modality, *and duration* per sub-action—grows exponentially with the number of sub-actions, rendering exhaustive search impractical.

These challenges motivate the online, frame-based Lyapunov-guided policy search developed in Sec. IV, which decouples the long-term constraints (34e)–(34f) into per-frame decisions via a ratio form of drift-plus-penalty, and employs a generative world-action model (Sec. III) both to propose a small set of high-quality candidate chunks and to score them by rolling out the coupled perception–motion dynamics.

III. WORLD-ACTION MODEL

To score a candidate chunk before it is executed, the controller maintains a world-action model composed of an on-

vehicle uncertainty evaluator and a BS-side generative model, which instantiate the ego and environment experts of Sec. I. Both are trained offline and only queried at run time.

A. BS-side Generative Model

Scoring a candidate therefore requires predicting how the joint (ego, environment, observation) state evolves. Conditioned on the global context $\mathcal{C}_{v,t_k}^{\text{BS}}$, the generative model $\mathcal{W}_\theta$ proposes a small set of candidate chunks and rolls out the future observations each would produce.

For a candidate $\mathbf{a} = (a^{(0)}, \dots, a^{(J-1)})$ with planned horizon $F(\mathbf{a})$, the controller rolls out the coupled dynamics over that horizon. Starting from $\mathcal{C}_{v,t_k}^{\text{BS}}$, for each step $\tau = 0, \dots, F(\mathbf{a}) - 1$ with active sub-action $a^{(j(\tau))}$, where $j(\tau)$ is the unique index such that $t_k + \tau \in \mathcal{E}_k^{(j(\tau))}$: (i) $\mathcal{U}_\phi$ evaluates the current rolled observation to obtain the predicted per-slot uncertainty $\hat{U}_{v,t_k+\tau}(\mathbf{a})$ via (16); (ii) the risk-aware controller (32) selects u^* and advances the ego; (iii) $\mathcal{W}_\theta$ predicts the next observation under $a^{(j(\tau))}$, accounting for member deliveries with their latency and freshness, and the predicted payload arrivals $\hat{L}_m(t_k + \tau)$ and service rates $\hat{R}_{m,t_k+\tau}$ that enter the queue terms of the per-frame program (41). The rollout thus returns, for each candidate, the predicted uncertainty trajectory $\{\hat{U}_{v,t_k+\tau}(\mathbf{a})\}_{\tau=0}^{F(\mathbf{a})-1}$ together with the predicted communication load; the trajectory of the installed chunk is stored as the reference $\{\hat{U}_{v,t|t_k}\}$ used by the trigger in (8). Note that both the bandwidth trajectory $\{B_m^{(j)}\}$ and the durations $\{n^{(j)}\}$ of a candidate are decision variables rather than predictions, so the budget-related terms and the horizon normalization in (41) are evaluated exactly; only the uncertainty and queue arrival/service terms rely on the rollout.

IV. ALGORITHM

We propose an online solution that applies frame-based Lyapunov drift-plus-penalty in ratio form to decouple the long-term constraints (34e)–(34f) and reduce **(P1)** to a per-frame static program that depends only on the current system state. We refer to the resulting scheme as *Lyapunov-guided world-action-model policy search*.

A. Virtual queue for the bandwidth budget

To enforce the long-term budget (34e) without a hand-tuned trade-off weight, we introduce a virtual queue $Z(t)$ with $Z(0) = 0$ and update

$$Z(t+1) = \max \left\{ Z(t) + \underbrace{\sum_{m \in \mathcal{S}_t} B_{m,t} - \bar{B}_{\text{bgt}}}_{\triangleq x(t)}, 0 \right\}, \quad (35)$$

which accumulates the historical budget overrun: bandwidth consumed beyond the per-slot budget allowance enters the queue, while unused allowance drains it. By the standard argument [1], mean-rate stability of $Z(t)$ implies $\bar{B} \leq \bar{B}_{\text{bgt}}$, so (34e) is transformed into a queue-stability requirement of the same type as (34f). Since $0 \leq \sum_m B_{m,t} \leq B_{\text{max}}$, the per-slot change of Z is bounded, $|x(t)| \leq \delta_Z \triangleq \max\{\bar{B}_{\text{bgt}}, B_{\text{max}} - \bar{B}_{\text{bgt}}\}$.

B. Frame-based drift-plus-penalty reduction

Let $\Theta(t) = \{Q_m(t)\}_m \cup \{Z(t)\}$ collect the member transmit backlogs (27) and the virtual queue (35), and define the quadratic Lyapunov function

$$\mathcal{L}(\Theta(t)) = \frac{1}{2} \sum_m Q_m(t)^2 + \frac{1}{2} Z(t)^2. \quad (36)$$

1) *Per-slot bound*: Squaring (27) with the inequality $(\max\{q - \mu, 0\} + A)^2 \leq q^2 + \mu^2 + A^2 + 2q(A - \mu)$ for $q, \mu, A \geq 0$, and squaring (35) with $(\max\{z + x, 0\})^2 \leq (z + x)^2$, we obtain, for an *arbitrary* (possibly slot-varying) action sequence and every slot t ,

$$\begin{aligned} \mathcal{L}(\Theta(t+1)) - \mathcal{L}(\Theta(t)) &\leq \Omega_0 + \sum_m Q_m(t) (L_m(t) - R_{m,t} T_s) \\ &\quad + Z(t) x(t), \end{aligned} \quad (37)$$

where $\Omega_0 = \frac{M}{2} (L_{\text{max}}^2 + (R_{\text{max}} T_s)^2) + \frac{1}{2} \delta_Z^2$ and M is the number of member links. By the single-member constraint (34c), at most one member is active per slot; a de-selected member receives no arrival and is granted no bandwidth ($L_m(t) = 0, R_{m,t} = 0$), so its queue is frozen and drops out of (37). The bound uses only the slot- t update and therefore holds regardless of the selection at any other slot—a property we exploit below when frames are truncated by the mismatch trigger.

2) *Frame telescoping and price freezing*: Summing (37) over the frame $\mathcal{F}_k$ yields queue weights $Q_m(t_k + \tau)$ and $Z(t_k + \tau)$ that evolve *inside* the frame, whereas an implementable per-frame decision may depend only on the state observed at the frame boundary. We therefore *freeze the prices* at t_k : since each queue changes by at most δ_{max} (resp. δ_Z) per slot,

$$\begin{aligned} Q_m(t_k + \tau) (L_m - R_{m,t_k} T_s) &\leq Q_m(t_k) (L_m - R_{m,t_k} T_s) + \tau \delta_{\text{max}}^2, \\ Z(t_k + \tau) x &\leq Z(t_k) x + \tau \delta_Z^2. \end{aligned} \quad (38)$$

Summing (37) with (38) over $\tau = 0, \dots, F_k - 1$ and bounding $F_k \leq F_{\text{max}}$ gives the frame bound

$$\begin{aligned} \mathcal{L}(\Theta(t_{k+1})) - \mathcal{L}(\Theta(t_k)) &\leq \Omega_1 \\ &\quad + \sum_m Q_m(t_k) \sum_{\tau=0}^{F_k-1} (L_m(t_k + \tau) - R_{m,t_k+\tau} T_s) \\ &\quad + Z(t_k) \sum_{\tau=0}^{F_k-1} x(t_k + \tau), \end{aligned} \quad (39)$$

with

$$\Omega_1 = F_{\text{max}} \Omega_0 + \frac{F_{\text{max}}^2}{2} (M \delta_{\text{max}}^2 + \delta_Z^2). \quad (40)$$

The $\mathcal{O}(F_{\text{max}}^2)$ scaling of Ω_1 quantifies the theoretical price of committing to a longer plan without re-pricing at sub-epoch boundaries, and provides one of the considerations bounding the horizon cap F_{max} (the other being the compounding rollout error of the world model).

3) *Per-frame program in ratio form*: Adding the Λ -weighted frame-accumulated penalty—which, per the objective (33), comprises the per-slot uncertainty *plus the per-frame planning overhead* c_0 , the bandwidth being accounted for through Z —to (39) yields the frame-level drift-plus-penalty expression. Because the planned horizon now varies across candidates, minimizing the frame *sum* would be systematically biased: the per-slot terms are essentially nonnegative, so a shorter chunk always yields a smaller sum, and the planner would degenerate to the maximal re-planning rate—defeating the purpose of chunking. The quantity that the long-run average (33) actually accumulates is the cost *per unit time*; by the renewal-reward structure of the frame process [1], the correct per-frame criterion is therefore the ratio of the frame cost to the frame length. Since the planned horizon $F(\mathbf{a})$ is a decision variable known exactly at decision time, the ratio is well defined without any prediction of the frame length, and the per-frame program is

$$\begin{aligned}
(\mathbf{P2}) \quad \mathbf{a}_{t_k}^* = \arg \min_{\mathbf{a} \in \mathcal{A}_{\text{cand}}} \quad & \frac{1}{F(\mathbf{a})} \left\{ \Lambda c_0 + \sum_{\tau=0}^{F(\mathbf{a})-1} \left[\Lambda \hat{U}_{v,t_k+\tau}(\mathbf{a}) \right. \right. \\
& + Z(t_k) \left(\sum_{m \in \mathcal{S}^{(j(\tau))}} B_m^{(j(\tau))} - \bar{B}_{\text{bgt}} \right) \\
& \left. \left. + \sum_m Q_m(t_k) (\hat{L}_m(t_k+\tau) - \hat{R}_{m,t_k+\tau} T_s) \right] \right\} \\
\text{s.t.} \quad & |\mathcal{S}^{(j)}| \leq 1, \quad j = 0, \dots, J-1, \\
& 0 \leq \sum_{m \in \mathcal{S}^{(j)}} B_m^{(j)} \leq B_{\max}, \quad j = 0, \dots, J-1, \\
& n^{(j)} \in \mathbb{N}, \quad n^{(j)} \geq n_{\min}, \quad j = 0, \dots, J-1, \\
& F(\mathbf{a}) \leq F_{\max}.
\end{aligned} \tag{41}$$

where $j(\tau)$ indexes the sub-epoch containing slot $t_k + \tau$, $\mathcal{A}_{\text{cand}}$ is the candidate set proposed by $\mathcal{W}_\theta$. Note that the term $-Z(t_k)\bar{B}_{\text{bgt}}$ becomes candidate-independent after division by $F(\mathbf{a})$ and may be dropped in implementation. **(P2)** depends only on the current backlogs $\{Q_m(t_k)\}$, the virtual queue $Z(t_k)$, and the current context. Because the per-slot bound (37) holds for any action sequence, the BS evaluates (41) only at each epoch and holds the minimizer over $[t_k, t_{k+1})$; re-optimization occurs at the event-driven epochs (8), while the stability guarantees of Proposition 1 are retained.

C. Cost-rate interpretation and adaptive prices

Dividing through, the objective of (41) decomposes into an interpretable *cost rate*:

$$\begin{aligned}
g(\mathbf{a}) = & \underbrace{\frac{\Lambda c_0}{F(\mathbf{a})}}_{\text{amortized planning cost}} + \underbrace{\frac{\Lambda}{F(\mathbf{a})} \sum_{\tau} \hat{U}_{v,t_k+\tau}(\mathbf{a})}_{\text{uncertainty rate}} \\
& + Z(t_k) \underbrace{\frac{1}{F(\mathbf{a})} \sum_{\tau} \sum_m B_m^{(j(\tau))}}_{\text{bandwidth rate}} \\
& + \sum_m Q_m(t_k) \underbrace{\frac{1}{F(\mathbf{a})} \sum_{\tau} (\hat{L}_m(t_k+\tau) - \hat{R}_{m,t_k+\tau} T_s)}_{\text{net-load rate}},
\end{aligned} \tag{42}$$

up to the candidate-independent constant $-Z(t_k)\bar{B}_{\text{bgt}}$. Three *adaptive prices* govern the trade-offs, none of which is hand-tuned. The link backlog $Q_m(t_k)$ is a per-link congestion price: a backlogged link is admitted only if it can be drained (high rate, hence more bandwidth), while an empty queue lets the Λ -weighted uncertainty rate dominate. The virtual queue $Z(t_k)$ is the spectrum-budget price: when recent consumption has been frugal (Z small), bandwidth is nearly free and the optimizer spends it aggressively to suppress uncertainty; after sustained overruns (Z large), bandwidth becomes expensive and only cooperation with a large uncertainty payoff is admitted. In the long run $Z(t)$ converges to the level at which the budget constraint is met, i.e., it is an online estimate of the shadow price of (34e): **the trade-off weight that a scalarized formulation would require as a manually tuned input is instead learned by the system.** Finally, the amortized term $\Lambda c_0/F(\mathbf{a})$ prices the decision cadence itself: in quasi-stationary conditions the rollout foresees little benefit from reconfiguration, the remaining rates are insensitive to the horizon, and the amortization drives the planner toward long frames; when the rollout foresees uncertainty growth under a stale configuration (e.g., a member about to leave the corridor), shorter durations become worthwhile despite the higher amortized overhead. The decision-cost-versus-adaptivity trade-off, previously a narrative motivation for chunking, is thereby internalized in the objective. Two consequences of retaining the maximal-duration local-only chunk as an always-feasible candidate are worth noting: *whether* to cooperate at all is decided endogenously—a cooperative chunk is installed only if its price-adjusted cost rate beats the local baseline—and the installed chunk’s predicted cost rate never exceeds that baseline, a no-degradation property that holds by optimality.

D. Online procedure and performance

Algorithm 1 summarizes the operation from registration onward. At each epoch the BS solves (41), installs $\mathbf{a}_{t_k}^*$, and stores its reference uncertainty trajectory; between epochs the real and virtual queues evolve under (27) and (35) while the vehicle monitors the realized uncertainty every T_a and reports mismatch events.

Algorithm 1 Lyapunov-guided world-action-model policy search

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1: Input: weight  $\Lambda$ , mismatch tolerance  $\epsilon_{\text{gap}}$ , budget  $\bar{B}_{\text{bgt}}$ ,
   planning cost  $c_0$ , chunk size  $J$ , duration bound  $n_{\text{min}}$ ,
   horizon cap  $F_{\text{max}}$ , dwell  $T_{\text{min}}$ , monitoring period  $T_a$ ;
   world-action model  $(\mathcal{W}_\theta, \mathcal{U}_\phi)$ 
2: On registration: initialize  $Q_m \leftarrow 0 \forall m$ ,  $Z \leftarrow 0$ ; install a
   local-only chunk; set  $t_0 \leftarrow$  registration time
3: for each epoch  $t_k$  do
4:   BS forms context  $\mathcal{C}_{v,t_k}^{\text{BS}}$  and reads  $\{Q_m(t_k)\}$ ,  $Z(t_k)$ 
5:    $\mathcal{A}_{\text{cand}} \leftarrow$  propose candidate chunks (members, band-
     widths, modalities, durations) from  $\mathcal{W}_\theta$ , plus the maximal-
     duration local-only chunk
6:   for each candidate  $\mathbf{a} \in \mathcal{A}_{\text{cand}}$  do
7:     Roll out the coupled dynamics over
      $\tau = 0, \dots, F(\mathbf{a})-1$  with (32); obtain  $\{\hat{U}_{v,t_k+\tau}(\mathbf{a})\}$ ,
      $\{\hat{L}_m, \hat{R}_m\}$ 
8:     Optimize  $\{B_m^{(j)}\}$  and evaluate the cost rate (41)
9:   end for
10:   $\mathbf{a}_{t_k}^* \leftarrow$  minimizer of (41); deliver it to the vehicle; store
    the reference trajectory  $\{\hat{U}_{v,t|t_k}\}$ 
11:  while  $t < t_k + F(\mathbf{a}_{t_k}^*)$  do
12:    Advance one slot: update  $\{Q_m(t)\}$  via (27) and
     $Z(t)$  via (35)
13:    Every  $T_a$ : vehicle computes realized  $U_{v,t}$  and
    reports the deviation  $U_{v,t} - \hat{U}_{v,t|t_k}$ 
14:    if  $t > t_k + T_{\text{min}}$  and  $|U_{v,t} - \hat{U}_{v,t|t_k}| > \epsilon_{\text{gap}}$  then
15:      break  $\triangleright$  plan invalidated; trigger re-planning
16:    end if
17:  end while
18:   $t_{k+1} \leftarrow t$ 
19: end for

```

The minimum dwell T_{min} in (8) bounds the re-planning rate under repeated mismatch and, as shown below, lower-bounds the expected frame length appearing in the optimality gap. The performance guarantee rests on the following assumptions.

Assumption 1 (Boundedness). *Payloads, rates, and bandwidth allocations are bounded, so that $L_m(t) \leq L_{\text{max}}$, $R_{m,t} \leq R_{\text{max}}$, and $|x(t)| \leq \delta_Z$ for all t .*

Assumption 2 (Stationary benchmark). *The environment state evolves ergodically, and there exists a stationary randomized policy that attains the optimal objective $U_c^{\text{opt}}(\bar{B}_{\text{bgt}})$ of (P1) while satisfying (34e)–(34f). Moreover, the Slater condition holds: the local-only policy satisfies the budget constraint with margin $\epsilon_s = \bar{B}_{\text{bgt}} > 0$, since it consumes zero bandwidth.*

Assumption 3 (Ω_w -additive approximation). *At every epoch, the chunk returned by the world-action-model search attains the objective of (41) within a bounded additive error $\Omega_w \geq 0$, which accounts for the finite candidate set, the rollout prediction error, and the mismatch-truncation effect.*

Proposition 1 (Budget compliance, stability, and optimality gap). *Suppose Assumptions 1–3 hold and $\bar{B}_{\text{bgt}} > 0$. Then*

Algorithm 1 guarantees:

- (i) *all member queues $Q_m(t)$ and the virtual queue $Z(t)$ are mean-rate stable; consequently the long-term budget constraint (34e) and the stability constraint (34f) are satisfied;*
- (ii) *the overhead-inclusive long-term average cost satisfies*

$$\bar{U}_c \leq U_c^{\text{opt}}(\bar{B}_{\text{bgt}}) + \frac{\Omega_1 + \Omega_w}{\Lambda \mathbb{E}[F]}, \quad (43)$$

with $T_{\text{min}} \leq \mathbb{E}[F] \leq F_{\text{max}}$ and Ω_1 given in (40);

- (iii) *the time-average total backlog obeys*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left[\sum_m Q_m(t) + Z(t) \right] = \mathcal{O} \left(\frac{\Omega_1 + \Omega_w + \Lambda (F_{\text{max}} + c_0)}{\bar{B}_{\text{bgt}}} \right), \quad (44)$$

where we used $U_{v,t} \leq 1$ so that the frame cost is at most $\Lambda(F_{\text{max}} + c_0)$, and the finite-horizon budget overrun decays as $Z(T)/T = \mathcal{O}(\Lambda/T)$.

Proof sketch. Adding the Λ -weighted frame penalty (including the per-frame overhead Λc_0) to (39) and invoking Assumption 3, the chunk installed by Algorithm 1 attains a frame cost rate within $(\Omega_1 + \Omega_w)$ -additive error of that achieved by any alternative, in particular by the stationary randomized policy of Assumption 2 evaluated under the same frame structure (the trigger rule applies identically to both, and the local-only chunk keeps the comparison well defined). Summing the resulting inequality over N frames, telescoping the Lyapunov terms, dividing by the expected total time $N \mathbb{E}[F]$, and letting $N \rightarrow \infty$ yields (ii) by the renewal-reward theorem—the per-epoch cost c_0 contributing the term $c_0/\mathbb{E}[F]$ that matches the epoch-counting term of (33)—and (i), (iii) by the standard Lyapunov stability argument with the Slater margin $\epsilon_s = \bar{B}_{\text{bgt}}$ supplied by the local-only policy; see the frame-based drift-plus-penalty analysis in [1]. The bound $Z(T) \leq Z(0) + \mathcal{O}(\Lambda)$ follows from (iii) and gives the finite-horizon overrun rate. $\square$

Proposition 1 exhibits the classical $[\mathcal{O}(1/\Lambda), \mathcal{O}(\Lambda)]$ utility–backlog trade-off with two features specific to our setting. First, the constant $\Omega_1 = \mathcal{O}(F_{\text{max}}^2)$ grows quadratically in the horizon cap, so F_{max} trades foresight against both the price-freezing constant and the compounding rollout error, and is a design parameter to be selected moderately rather than maximized. Second, the expected frame length $\mathbb{E}[F]$ in the denominator of (43) is now shaped from two sides: it is *chosen* by the planner through the durations—with the amortized cost $\Lambda c_0/F$ favoring long frames in quasi-stationary conditions—and *truncated* by realized model mismatch, with the dwell time T_{min} guarding against pathological high-frequency truncation and a better-calibrated world model enlarging $\mathbb{E}[F]$ (Remark ??), directly tightening the gap. The remaining algorithm parameter Λ controls only the convergence–backlog trade-off; the perception–communication trade-off itself is set by the physically meaningful budget $\bar{B}_{\text{bgt}}$ and priced online by $Z(t)$, while the decision cadence is priced by the calibrated planning cost c_0 .

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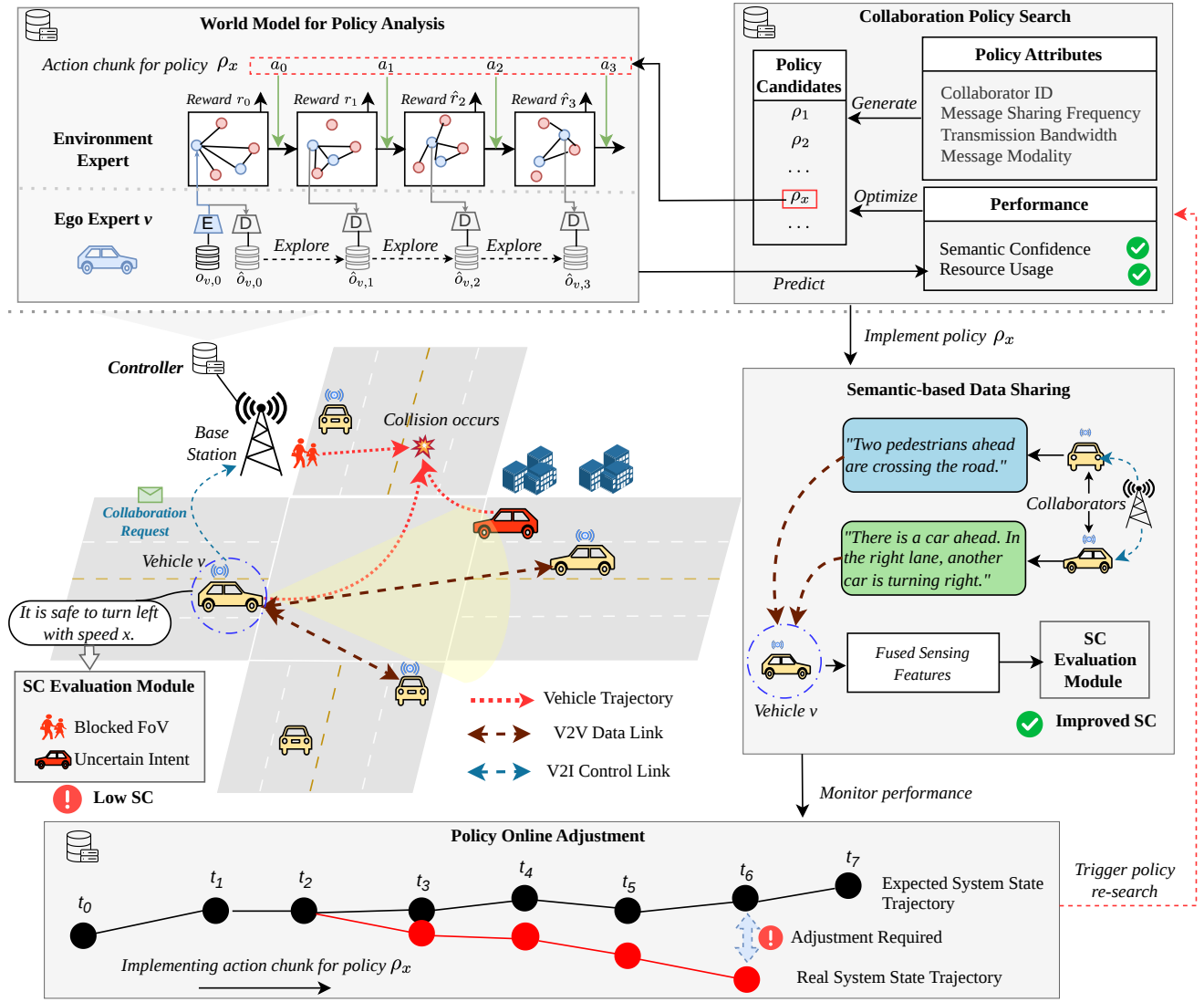


Fig. 1. Scenario

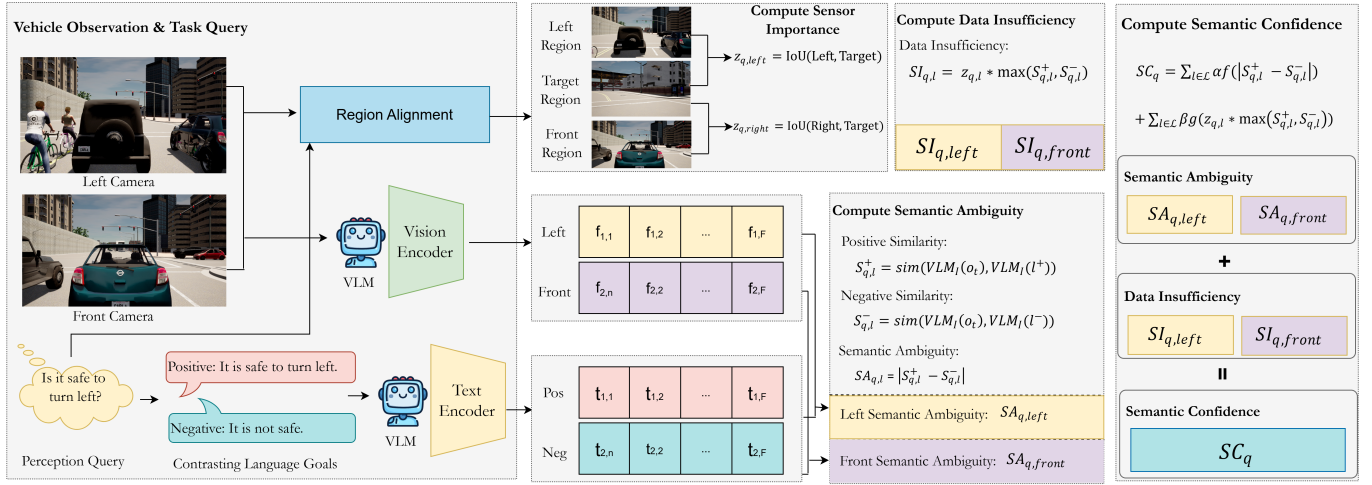


Fig. 2. Semantic Confidence